



Department of Computer Science

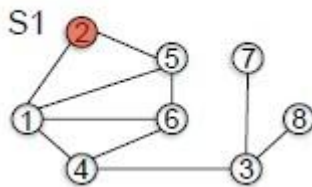
National University of Computer and Emerging Sciences
Lahore Campus

Data Mining

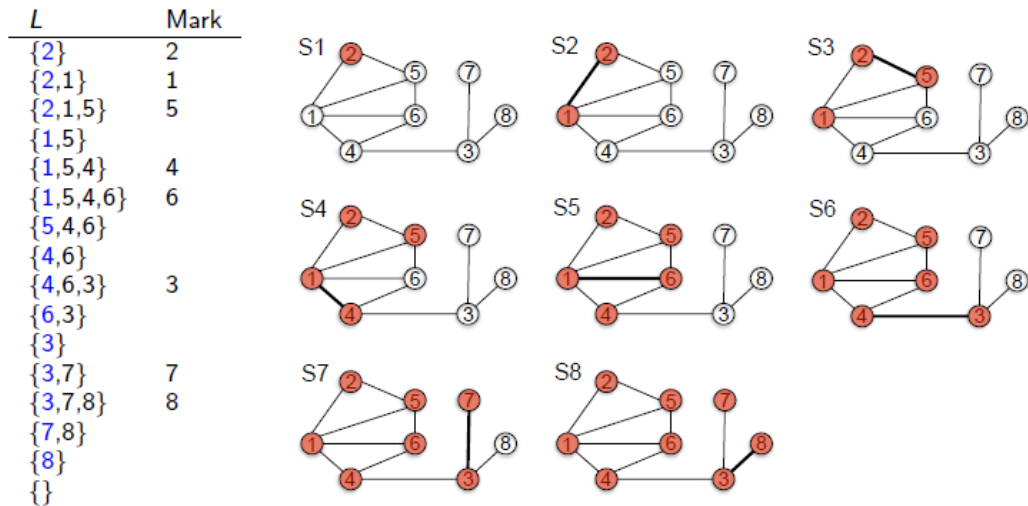
Section A

Quiz 3b

- 1) Traverse following graph using BFS by mentioning L and M lists at each step (4 marks)

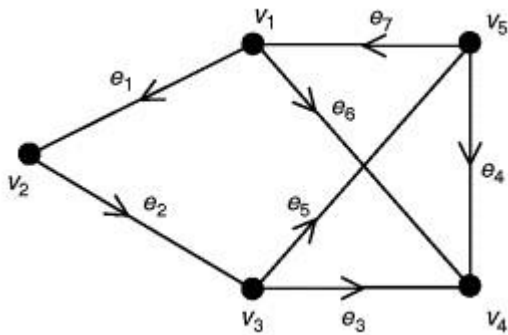


► Below we indicate the **chosen** and **marked** nodes. Initialize $s = 2$



► The algorithm builds a wider tree (breadth first)

2) Find incidence matrix for following directed graph. (2 marks)



$= (-1, 0, 0, 0, 0, -1, 1$
 $1, -1, 0, 0, 0, 0, 0$
 $0, 1, -1, 0, 1, 0, 0$
 $0, 0, 1, 1, 0, 1, 0$
 $0, 0, 0, -1, 1, 0, -1)$

3) Calculate graph density of above graph (1 mark)

$$= \frac{Ne}{Nv^2}$$

$$= \frac{7}{(5^2)}$$

$$= \frac{7}{25}$$

4) Find in and out degree of each vertex from above graph (1.5 mark)

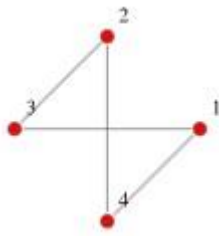
V1 $d_{in} = 1, d_{out} = 2$

V2 $d_{in} = 1, d_{out} = 1$

V3 $d_{in} = 1, d_{out} = 2$

V4 $d_{in} = 3, d_{out} = 0$

V5 $d_{in} = 1, d_{out} = 2$



5)

Find the adjacency matrix for above graph. (1.5 marks)

$$= (0,0,1,1$$

$$0,0,1,1$$

$$1,1,0,0$$

$$1,1,0,0)$$

